OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Chemistry

Aldehydes, Ketones & Carboxylic Acids

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Master Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

www.ourmocks.com

Syllabus Alignment & CUET UG Exam Blueprint

Sub-topic	Official Syllabus Parameters (NTA CUET UG Baseline)	Core Exam Tested Skills
Nomenclature & Carbonyl Structure	Nomenclature of aldehydes, ketones, and carboxylic acids; nature of carbonyl group ($C=O$ polarity, sp^2 hybridization, dipole moments, planar geometry).	Identifying IUPAC names, explaining Bürgi-Dunitz trajectory, and ranking electrophilic reactivity based on steric crowding and inductive effects.
Methods of Preparation	Preparation of aldehydes and ketones from alcohols (PCC, CrO_3), dehydrogenation, ozonolysis of alkenes, hydration of alkynes, Rosenmund reduction, Stephen reaction, Etard reaction, Gattermann-Koch reaction, and Friedel-Crafts acylation.	Selecting poisoned catalyst systems ($Pd/BaSO_4$), differentiating chromyl chloride complexes, and predicting hydration regiochemistry with $\frac{Hg^{2+}}{H^+}$.
Physical & Chemical Properties	Physical trends (boiling points, water solubility via H-bonding). Nucleophilic addition mechanisms (HCN, $NaHSO_3$, $R-MgX$, alcohols, and ammonia derivatives $G-NH_2$).	Evaluating nucleophilic addition rates ($HCHO > RCHO > R_2$ CO), identifying semicarbazide nucleophilic sites, and explaining reversible bisulphite adduct purification.
Hallmark Reactions & Oxidation/Reduction	Reactivity of α-hydrogens: Aldol condensation, Cross-aldol condensation, Cannizzaro reaction, Cross-Cannizzaro. Clemmensen reduction (Zn-Hg/HCl) and Wolff-Kishner reduction ($NH_2\frac{NH_2}{KOH}$).	Predicting aldol products vs Cannizzaro disproportionation (presence vs absence of α-H), choosing acid-stable vs base-stable deoxygenation reagents.
Diagnostic Distinction Tests	Tollens' silver mirror test, Fehling's test, Iodoform reaction ($\frac{I_2}{NaOH}$), Schiff's test, and 2,4-DNP Brady's reagent.	Distinguishing aliphatic vs aromatic aldehydes (Fehling's failure on benzaldehyde), recognizing the $CH_3-C{=}O$ and $CH_3-CH(OH)$- motifs.
Carboxylic Acids & Acidity Architecture	Nomenclature, structure of carboxyl group, preparations (nitrile hydrolysis, Grignard with CO_2, primary alcohol oxidation). Acidity, carboxylate resonance, inductive effects, Hell-Volhard-Zelinsky (HVZ) reaction, decarboxylation with soda-lime.	Explaining carboxylate equivalent resonance, ranking halogenated acids and substituted benzoic acids (pK_a trends), identifying α-halogenation conditions.

EXAM PATTERN

CUET UG Paper Pattern & Weightage: The Chemistry paper carries **50 compulsory questions** in **60 minutes** (+5 marks for correct, −1 mark for incorrect). Unit VIII (Aldehydes, Ketones and Carboxylic Acids) is the highest-weight organic chemistry unit, contributing 6 to 8 questions focusing on Rosenmund catalyst poisoning, Aldol/Cannizzaro discriminating rules, Tollens'/Fehling's/Iodoform tests, and Carboxylic acid pK_a substituent orders.

Module 1: Nomenclature, Carbonyl Geometry & Molecular Polarity

1. Electronic Structure of the Carbonyl ($C = O$) Group

1. **Orbital Hybridization & Bond Geometry:**
   - The carbonyl carbon is **sp^2-hybridised** and forms three coplanar σ-bonds at approximately 120° angles with planar trigonal geometry.
   - The unhybridised $2p$ orbital of carbon overlaps laterally with a parallel $2p$ orbital of oxygen to construct a **π-bond**.
   - Oxygen retains two non-bonding lone pairs housed in its remaining sp^2 hybrid orbitals.
2. **Dipole Moment & Polar Character:**
   - Oxygen is markedly more electronegative (3.44) than carbon (2.55). Consequently, the π-electron cloud is drawn heavily toward oxygen, generating substantial resonance polarization:

$$> C = O \longleftrightarrow > C^{+} - O^{-}$$

   - The carbonyl carbon functions as an **electrophilic (Lewis acid) centre**, whereas the carbonyl oxygen acts as a **nucleophilic (Lewis base) centre**.
   - Carbonyl compounds possess high dipole moments ($\mu \approx 2.3 - 2.8$ D), significantly greater than corresponding haloalkanes or ethers.

2. IUPAC Nomenclature & Structural Isomerism

- **Aldehydes ($R - CHO$):** The carbonyl carbon is universally assigned position C-1; designated by suffix **-al** (e.g. methanal, ethanal, 2-methylbutanal, benzaldehyde or benzenecarbaldehyde).
- **Ketones ($R - CO - R'$):** Named by identifying the longest continuous carbon chain containing the $C = O$ carbon, numbered to give the carbonyl carbon the lowest locant; suffix **-one** (e.g. propan-2-one, butan-2-one, acetophenone or 1-phenylethan-1-one).
- **Carboxylic Acids ($R - COOH$):** Suffix **-oic acid** (e.g. methanoic acid, ethanoic acid, benzoic acid, benzene-1,2-dicarboxylic acid / phthalic acid).

3. Physical Properties: Polarity vs Hydrogen Bonding

1. **Boiling Point Progression:**

$$\text{Alkanes} < \text{Ethers} < \text{Aldehydes \& Ketones} < \text{Alcohols} < \text{Carboxylic Acids}$$

   - Aldehydes and ketones engage in strong **dipole-dipole interactions**, but **cannot form intermolecular hydrogen bonds among themselves** (lacking an $O - H$ proton).
   - Hence, their boiling points are higher than non-polar alkanes/ethers of comparable molecular mass, but lower than alcohols and carboxylic acids which exhibit extensive intermolecular H-bonding.

2. **Solubility in Water:**
   - Lower members (methanal, ethanal, propanone) are infinitely miscible with water due to hydrogen bonding between the **carbonyl oxygen** and water molecules:

$$> C = O^{\delta-} \cdots H^{\delta+} - O - H$$

   - As the non-polar hydrophobic hydrocarbon chain length extends beyond 4 carbons, solubility drops precipitously.

Module 2: Preparative Methods for Aldehydes & Ketones

1. Common Preparative Methods (Aldehydes & Ketones)

1. **Controlled Oxidation of Alcohols:**
   - **Primary (1°) Alcohols:** Selectively oxidised to aldehydes using **Pyridinium chlorochromate (PCC)** in dichloromethane (CH_2Cl_2) or Collin's reagent (CrO_3 in pyridine).

$$R - CH_2\ OH \xrightarrow{PCC} R - CHO$$

     CUET Exam Trap: Aqueous acidic dichromate ($K_2Cr_2\frac{O_7}{H_2}SO_4$) or permanganate ($KMnO_4$) overshoots aldehydes to yield carboxylic acids ($R - COOH$).
   - **Secondary (2°) Alcohols:** Oxidised smoothly to ketones using chromic anhydride (CrO_3) in glacial acetic acid or acidic dichromate:

$$R - CH(OH) - R' \xrightarrow{CrO_3} R - CO - R'$$

2. **Catalytic Dehydrogenation over Heated Copper ($\frac{Cu}{573}$ K):**
   - 1° Alcohol $\xrightarrow[573\ K]{Cu}$ Aldehyde $+ H_2 \uparrow$
   - 2° Alcohol $\xrightarrow[573\ K]{Cu}$ Ketone $+ H_2 \uparrow$
   - 3° Alcohol $\xrightarrow[573\ K]{Cu}$ Alkene (Dehydration; no dehydrogenation possible) $+ H_2\ O$
3. **Ozonolysis of Alkenes followed by Reductive Workup (Zn / H_2O):**
   - Ozonolysis cleaves the carbon-carbon double bond cleanly. Zinc dust destroys hydrogen peroxide (H_2O_2), preventing oxidation of aldehydes to carboxylic acids.
   - Symmetrically disubstituted alkenes ($R - CH{=}CH - R$) yield 2 equivalents of aldehyde; tetrasubstituted alkenes yield ketones.
4. **Hydration of Alkynes ($\frac{HgSO_4}{H_2}SO_4$ at 333 K):**
   - Addition of water across triple bonds occurs via Markovnikov regioselectivity:
   - **Ethyne (CH ≡ CH):** The ONLY alkyne that yields an **aldehyde (Ethanal)** via vinyl alcohol tautomerisation:

$$HC \equiv CH + H_2\ O \xrightarrow[H^+]{Hg^{2+}} [CH_2 = CH\text{-}OH] \leftrightarrow CH_3 - CHO$$

   - **All other terminal and internal alkynes:** Yield **Ketones** (e.g. propyne gives propan-2-one).

2. Exclusive Preparative Methods for Aldehydes

1. **Rosenmund Reduction (Acid Chloride to Aldehyde):**
   - Acyl chlorides are catalytically hydrogenated over **Palladium supported on Barium Sulphate (Pd / $BaSO_4$)** poisoned with sulphur or quinoline:

$$R-\mathrm{COCl}+\mathrm{H}_2 \xrightarrow{\text{Pd / } \mathrm{BaSO_4}\text{/ S, quinoline}} R-\mathrm{CHO}+\mathrm{HCl}$$

- **Poisoning Rationale:** Barium sulphate and sulphur reduce the catalytic activity of palladium, selectively stopping the reduction at the aldehyde stage and preventing over-reduction to primary alcohol.
- **Limitation:** Formaldehyde (HCHO) cannot be prepared by this method because formyl chloride (HCOCl) is unstable at room temperature.

2. **Stephen Reaction (Reduction of Nitriles):**
   - Nitriles are reduced with **stannous chloride and hydrochloric acid ($\mathbf{SnCl_2 + HCl}$)** in dry ether to yield aldimine hydrochloride, which on acidic hydrolysis yields aldehydes:

$$R-\mathrm{CN}+2[\mathrm{H}] \xrightarrow{\mathrm{SnCl_2}\text{/ HCl}} R-\mathrm{CH{=}NH}\cdot\mathrm{HCl} \xrightarrow{\mathrm{H_3O^+}} R-\mathrm{CHO}+\mathrm{NH_4\ Cl}$$

   - **Alternative Modern Reagent:** Diisobutylaluminium hydride (**DIBAL-H** or $\left[(\mathrm{CH_3})_2\mathrm{CHCH_2}\right]_2\ \mathrm{AlH}$) selectively reduces both nitriles and esters to aldehydes at 195 K without attacking double bonds.

3. **Etard Reaction (Toluene to Benzaldehyde):**
   - Oxidation of toluene with **chromyl chloride ($\mathbf{CrO_2Cl_2}$)** in carbon disulphide ($\mathrm{CS_2}$) or $\mathrm{CCl_4}$ yields a brown chromium complex, which on hydrolysis yields benzaldehyde:

$$\mathrm{C_6H_5-CH_3}+2\mathrm{CrO_2Cl_2} \rightarrow \mathrm{C_6H_5-CH(OCrOHCl_2)_2} \xrightarrow{\mathrm{H_3O^+}} \mathrm{C_6H_5-CHO}$$

4. **Gattermann-Koch Reaction (Direct Ring Formylation):**
   - Benzene reacts with carbon monoxide and hydrogen chloride in the presence of anhydrous aluminium chloride and cuprous chloride ($\frac{\mathrm{AlCl_3}}{\mathrm{CuCl}}$):

$$\mathrm{C_6H_6}+\mathrm{CO}+\mathrm{HCl} \xrightarrow{\text{anhy. } \mathrm{AlCl_3}\text{/ CuCl}} \mathrm{C_6H_5-CHO}+\mathrm{HCl}$$

3. Exclusive Preparative Methods for Ketones

1. **From Acyl Chlorides via Dialkylcadmium (R_2 Cd):**
   - Grignard reagents react with cadmium chloride to form dialkylcadmium, which reacts mildly with acid chlorides to yield ketones:

$$2R-\mathrm{MgX}+\mathrm{CdCl_2} \rightarrow R_2\ \mathrm{Cd}+2\ \mathrm{Mg(X)Cl}$$

$$2R'-\mathrm{COCl}+R_2\ \mathrm{Cd} \rightarrow 2R'-\mathrm{CO}-R+\mathrm{CdCl_2}$$

   Selectivity Factor: Dialkylcadmium is less nucleophilic than Grignard reagents and does NOT react further with the ketone product to form tertiary alcohols!

2. **Friedel-Crafts Acylation of Aromatic Rings:**
   - Benzene reacts with acetyl chloride or acetic anhydride in the presence of anhydrous aluminium chloride to yield **Acetophenone**:

$$\mathrm{C_6H_6}+\mathrm{CH_3\ COCl} \xrightarrow{\text{anhy. } \mathrm{AlCl_3}} \mathrm{C_6H_5-COCH_3}+\mathrm{HCl}$$

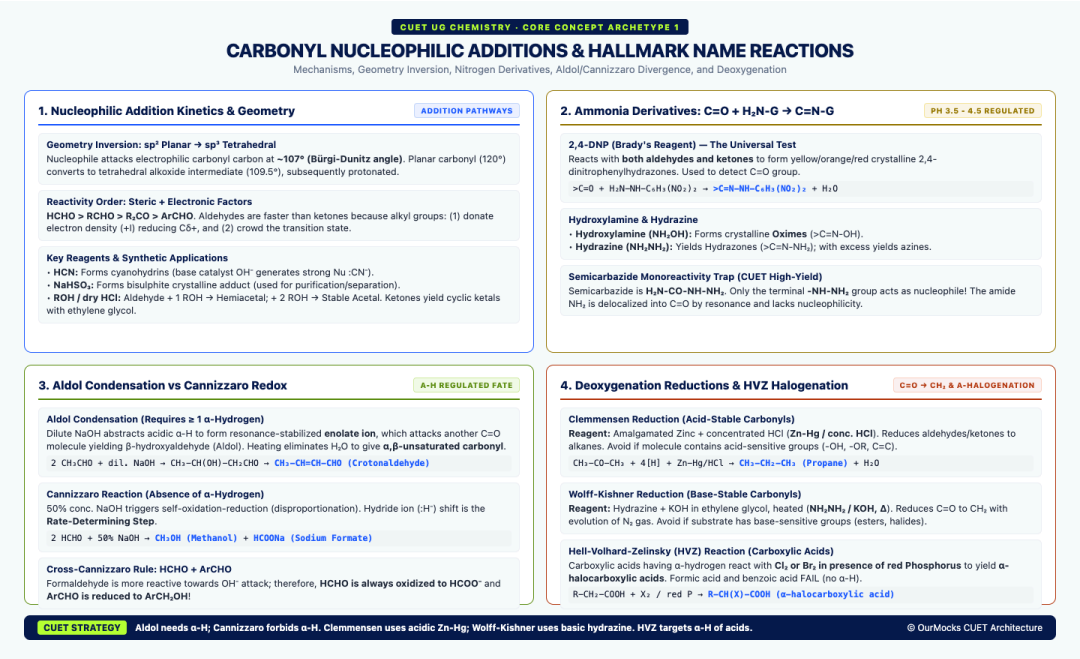


Figure 1: Carbonyl nucleophilic addition kinetics, orbital geometry changes, nitrogen derivatives (2,4-DNP, semicarbazone), Aldol vs Cannizzaro divergence, Clemmensen/Wolff-Kishner deoxygenation, and HVZ reaction.

Module 3: Nucleophilic Addition Reactions & Nitrogen Derivatives

1. Mechanism & Stereoelectronic Factors of Nucleophilic Addition

1. **Attack Geometry & Transition State:**
   - The attacking nucleophile approaches the planar sp^2 carbonyl carbon at the **Bürgi-Dunitz trajectory angle ($\approx 107^\circ$)**, displacing the π-electron pair onto oxygen.
   - The planar trigonal carbon (120°) transitions into a **tetrahedral alkoxide intermediate (109.5°)**, which abstracts a proton from the solvent to yield the neutral addition product.
2. **Reactivity Hierarchy (Aldehydes vs Ketones):**

$$\mathrm{HCHO} > \mathrm{CH_3\ CHO} > \mathrm{CH_3CH_2\ CHO} > \mathrm{CH_3COCH_3} > \mathrm{C_6H_5\ CHO} > \mathrm{C_6H_5COCH_3}$$

   - **Steric Factor:** Aldehydes possess only one alkyl group flanking the carbonyl carbon, allowing facile, unhindered nucleophilic attack. In ketones, two bulky alkyl groups severely crowd the transition state.
   - **Electronic Factor:** Alkyl groups are electron-donating ($+I$ inductive effect). In ketones, two alkyl groups pump electron density into the carbonyl carbon, heavily diminishing its partial positive charge ($\delta+$) and electrophilic attraction.
   - **Aromatic Effect:** In benzaldehyde ($\mathrm{C_6H_5\ CHO}$), the carbonyl carbon is resonance-conjugated with the benzene ring ($+R$ resonance donation), reducing its electrophilicity below aliphatic aldehydes.

2. Important Nucleophilic Addition Reactions

1. **Addition of Hydrogen Cyanide (HCN — Cyanohydrin Formation):**
   - Pure HCN adds extraordinarily slowly; the reaction is catalysed by base (OH^-) which generates the potent nucleophile cyanide ion ($:CN^-$):

$$> C = O + \mathrm{CN}^- \rightarrow > C(\mathrm{CN}) - O^- \xrightarrow{\mathrm{H}^+} > C(\mathrm{CN}) - \mathrm{OH} \quad \text{(Cyanohydrin)}$$

- Cyanohydrins are synthetic linchpins; acid hydrolysis converts the $-\mathrm{CN}$ group directly into an **α-hydroxy carboxylic acid**.

2. **Addition of Sodium Hydrogen Sulphite ($\mathrm{NaHSO_3}$):**
   - Carbonyl compounds react with saturated aqueous $\mathrm{NaHSO_3}$ to form water-soluble, white crystalline bisulphite addition adducts:

$$> C = O + \mathrm{NaHSO_3} \leftrightarrow > C(\mathrm{OH}) - \mathrm{SO_3^-Na^+} \quad \text{(Bisulphite adduct)}$$

   - **CUET Practical Application:** Because this reaction is readily reversible with dilute mineral acid or aqueous sodium hydroxide, it is employed universally to **separate and purify aldehydes and methyl ketones** from non-carbonyl organic mixtures.
3. **Addition of Alcohols (Hemiacetal and Acetal Formation):**
   - Aldehydes react with one equivalent of monohydric alcohol in the presence of dry HCl gas to form an alkoxyalcohol called a **Hemiacetal**:

$$R - \mathrm{CHO} + R' - \mathrm{OH} \overset{\text{dry HCl}}{\leftrightarrow} R - \mathrm{CH(OH)} - \mathrm{OR'} \quad \text{(Hemiacetal)}$$

   - Hemiacetals react with a second equivalent of alcohol to yield a gem-dialkoxy compound termed an **Acetal**:

$$R - \mathrm{CH(OH)} - \mathrm{OR'} + R' - \mathrm{OH} \overset{\text{dry HCl}}{\leftrightarrow} R - \mathrm{CH(OR')_2} + \mathrm{H_2\ O} \quad \text{(Acetal)}$$

   - Ketones do not react with simple monohydric alcohols under these conditions; they react smoothly with 1,2-diols (ethylene glycol) in dry HCl to yield stable cyclic **Ketals**.

3. Nucleophilic Addition-Elimination of Ammonia Derivatives ($G - \mathrm{NH_2}$)

Reaction general formula (acid-catalysed, optimum pH $3.5 - 4.5$):

$$> C = O + \mathrm{H_2\ N} - G \leftrightarrow [> C(\mathrm{OH}) - \mathrm{NH} - G] \xrightarrow{-\mathrm{H_2\ O}} > C = N - G$$

- **pH Sensitivity:** If the medium is too acidic ($\mathrm{pH} < 3$), the basic amine derivative protonates ($G - \mathrm{NH_3^+}$) and loses nucleophilicity. If too basic, carbonyl protonation fails.
- **Universal Identification — 2,4-DNP Test:** 2,4-Dinitrophenylhydrazine (Brady's Reagent) condenses with both aldehydes and ketones to produce bright **yellow, orange, or red crystalline 2,4-dinitrophenylhydrazones** with sharp melting points.
- **Semicarbazide Trap (CUET Favourite):** Semicarbazide is $\mathrm{H_2\ N - CO - NH - NH_2}$. It possesses two $-\mathrm{NH_2}$ groups. Only the **terminal hydrazine nitrogen ($\mathbf{-NH-NH_2}$)** acts as a nucleophile. The amide $-\mathrm{NH_2}$ is resonance-delocalized into the carbonyl group ($\mathrm{O{=}C - NH_2}$) and lacks nucleophilic reactivity!

Module 4: Hallmark Name Reactions & α-Hydrogen Reactivity

1. Acidity of α-Hydrogens in Carbonyl Compounds

The hydrogen atoms attached to the carbon adjacent to the carbonyl group (α-carbon) are unusually acidic ($pK_a \approx 19 - 20$) due to:

1. The powerful electron-withdrawing inductive effect ($-I$) of the carbonyl group polarizing the $\alpha - C - H$ bond.
2. The extraordinary resonance stabilization of the resulting conjugate base (**enolate ion**):

$$-C^{-}(\alpha)-\mathrm{C}{=}\mathrm{O} \longleftrightarrow -C(\alpha)=\mathrm{C}-\mathrm{O}^{-}$$

2. Aldol Condensation (Compounds with ≥ 1 α-Hydrogen)

- **Reaction:** Two molecules of an aldehyde or ketone containing at least one α-hydrogen undergo addition in the presence of dilute alkali (dil. NaOH, $\mathrm{Ba(OH)_2}$) to form a **β-hydroxyaldehyde (aldol)** or **β-hydroxyketone (ketol)**.
- **Dehydration:** Subsequent heating with acid or base causes spontaneous loss of water to yield an **α,β-unsaturated carbonyl compound**:

$$2\mathrm{CH_3\ CHO} \xrightarrow{\text{dil. NaOH}} \mathrm{CH_3-CH(OH)-CH_2\ CHO} \xrightarrow{\Delta,-\mathrm{H_2\ O}} \mathrm{CH_3-CH{=}CH\text{-}CHO} \quad \text{(But-2-enal / Crotonaldehyde)}$$

- **Cross-Aldol Condensation:** When condensation occurs between two different aldehydes or ketones, a mixture of four distinct products is formed if both possess α-hydrogens.
- If one carbonyl reactant lacks α-hydrogens (e.g. benzaldehyde + ethanal), only two products form, with benzalacetone / cinnamaldehyde as the predominant cross-aldol product.

3. Cannizzaro Reaction (Compounds with Zero α-Hydrogens)

- **Reaction:** Aldehydes that lack α-hydrogen atoms (e.g. Formaldehyde HCHO, Benzaldehyde $\mathrm{C_6H_5\ CHO}$, Trimethylacetaldehyde $\mathrm{(CH_3)_3\ C\text{-}CHO}$) do not undergo aldol condensation. When treated with **concentrated (50%) alkali (NaOH)**, they undergo **self-oxidation-reduction (disproportionation)**:

$$2\ \mathrm{HCHO} + \text{conc. NaOH} \rightarrow \mathrm{CH_3\ OH} + \mathrm{HCOONa} \quad \text{(Methanol + Sodium formate)}$$

$$2\mathrm{C_6H_5\ CHO} + \text{conc. NaOH} \rightarrow \mathrm{C_6H_5CH_2\ OH} + \mathrm{C_6H_5\ COONa} \quad \text{(Benzyl alcohol + Sodium benzoate)}$$

- **Mechanism Note:** Hydroxide attacks the carbonyl carbon to form a dianion; the **rate-determining step (RDS) is hydride ion (: H^-) transfer** from the intermediate to a second aldehyde molecule.
- **Cross-Cannizzaro Reaction:** When a mixture of benzaldehyde and formaldehyde is treated with concentrated alkali, formaldehyde is preferentially attacked by $\mathrm{OH^-}$ due to superior electrophilicity. Hence, **formaldehyde is exclusively oxidised to sodium formate (HCOONa)** while **benzaldehyde is exclusively reduced to benzyl alcohol ($\mathbf{C_6H_5CH_2\ OH}$)**.

4. Deoxygenation Reductions (Carbonyl $C=O \rightarrow \mathrm{CH_2}$)

1. **Clemmensen Reduction (Acid-Stable Carbonyls):**
   - Carbonyl group is reduced to methylene ($-\mathrm{CH_2}-$) using **amalgamated zinc and concentrated hydrochloric acid (Zn-Hg / conc. HCl)**:

$$>C=O + 4[\mathrm{H}] \xrightarrow{\text{Zn-Hg / conc. HCl}} > \mathrm{CH_2} + \mathrm{H_2\ O}$$

   - **Limitation:** Must NOT be used if the substrate contains acid-sensitive functional groups (such as $-\mathrm{OH}$, $-\mathrm{OR}$, or carbon-carbon double bonds).
2. **Wolff-Kishner Reduction (Base-Stable Carbonyls):**
   - Carbonyl compound is heated with **hydrazine ($\mathbf{NH_2NH_2}$)** followed by potassium hydroxide (KOH) in a high-boiling solvent (**ethylene glycol**, $453-473$ K):

$$>C=O + \mathrm{NH_2NH_2} \rightarrow >C=N-\mathrm{NH_2} \xrightarrow{\text{KOH / glycol},\Delta} > \mathrm{CH_2} + \mathrm{N_2}\uparrow$$

- **Limitation:** Must NOT be used if the substrate contains base-sensitive groups (such as halogen atoms or ester linkages).

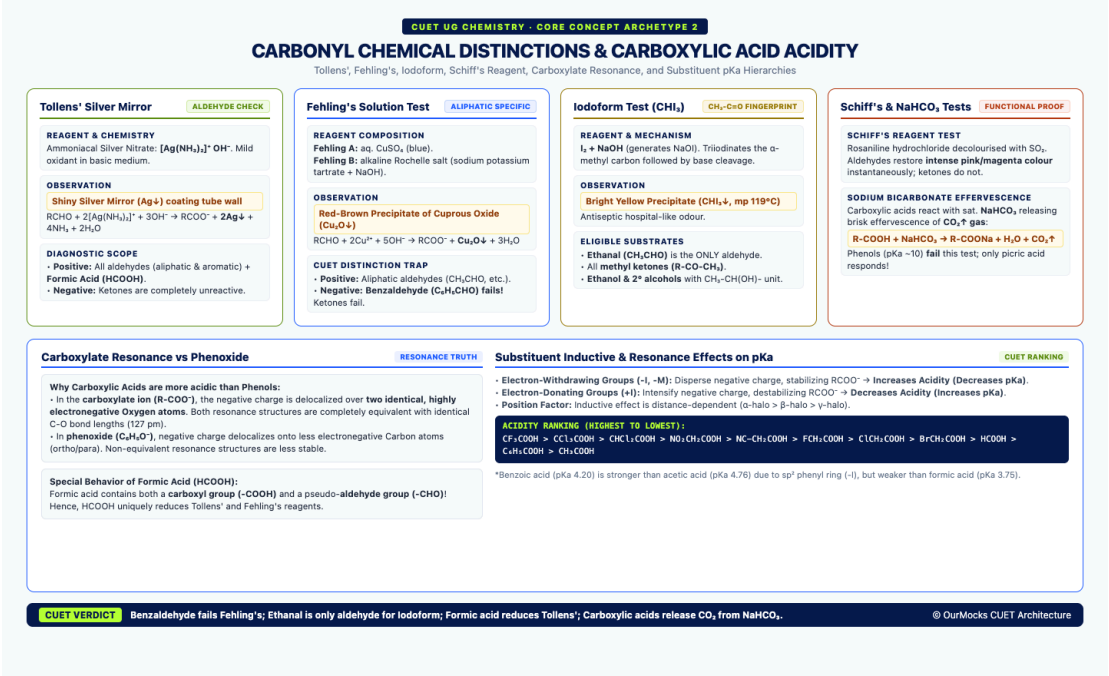


Figure 2: Diagnostic chemical tests (Tollens', Fehling's, Iodoform, Schiff's), carboxylate resonance equivalence, and substituent inductive effects on carboxylic acid pKa hierarchies.

Module 5: Diagnostic Distinction Tests & Functional Fingerprints

1. The Four Hallmark Carbonyl Tests

1. **Tollens' Test (Silver Mirror Test):**
   - **Reagent:** Ammoniacal silver nitrate solution ($\left[Ag(NH_3)_2\right]^+OH^-$), freshly prepared by adding NH_4 OH to $AgNO_3$ precipitate.
   - **Observation:** Formation of a sparkling **silver mirror** coating the inner glass tube wall, or a greyish-black silver precipitate:

$$R-CHO+2\left[Ag(NH_3)_2\right]^+ + 3OH^- \rightarrow R-COO^- + 2\ Ag\downarrow + 4NH_3 + 2H_2\ O$$

   - **Scope:** Positive for **ALL aldehydes** (both aliphatic and aromatic) and **Formic Acid (HCOOH)**. Ketones give a negative test.
2. **Fehling's Test (Aliphatic Aldehyde Check):**
   - **Reagent:** Equal volumes of Fehling A (aqueous $CuSO_4$, blue) and Fehling B (alkaline solution of sodium potassium tartrate / Rochelle salt).
   - **Observation:** Formation of a dense **red-brown precipitate of Cuprous Oxide (Cu_2 O)**:

$$R-CHO+2Cu^{2+}+5OH^- \rightarrow R-COO^- + Cu_2\ O\downarrow \quad \text{(Red-Brown ppt)} + 3H_2\ O$$

   - **CUET Critical Trap: Aliphatic aldehydes** give a positive Fehling's test. **Benzaldehyde and other aromatic aldehydes FAIL Fehling's test** because the carbonyl group is stabilized by ring resonance! Ketones also fail.
3. **Iodoform Test ($\frac{I_2}{NaOH}$ — Methyl Carbonyl Fingerprint):**
   - **Reagent:** Iodine in aqueous sodium hydroxide ($I_2 + NaOH$ -> generates sodium hypoiodite NaOI).

- **Observation:** Formation of a **bright yellow crystalline precipitate of Iodoform (CHI_3)** with a characteristic antiseptic odour (melting point 119° C):

$$R - COCH_3 + 3\ NaOI \rightarrow R - COONa + CHI_3 \downarrow \quad (\text{Yellow Crystals}) + 2\ NaOH$$

- **Scope:** Positive for compounds possessing either:
  1. A **methyl ketone group ($CH_3 - C{=}O$)**: Ethanal (the ONLY aldehyde), propanone, acetophenone, butan-2-one.
  2. A **methyl carbinol group ($CH_3 - CH(OH)$-)**: Ethanol, propan-2-ol, butan-2-ol (oxidised in situ to methyl ketones).

4. **Schiff's Test:**
   - Aldehydes restore the deep pink/magenta colour of Schiff's reagent (rosaniline hydrochloride decolourised by sulphur dioxide). Ketones give no restoration.

Module 6: Carboxylic Acids: Structure, Acidity & Reactions

1. Structure of Carboxyl Group & Extraordinary Acidity

1. **Carboxylate Resonance Equivalence:**
   - When a carboxylic acid ionises in water, it loses a proton to form a **carboxylate anion ($R - COO^-$)**:

$$R - COOH + H_2\ O \leftrightarrow R - COO^- + H_3O^+$$

   - In the carboxylate ion, the negative charge is delocalized over **two identical, highly electronegative oxygen atoms**. Both resonance structures are completely equivalent:

$$R - C(=O) - O^- \longleftrightarrow R - C(\text{-}\ O^-) = O$$

   - X-ray diffraction confirms that both carbon-oxygen bonds in the carboxylate ion are of identical length (127 pm), intermediate between single (136 pm) and double (122 pm) bonds.
   - In contrast, in the phenoxide ion ($C_6H_5O^-$), negative charge is delocalized onto less electronegative carbon atoms, giving unequal and less stable canonical structures.
   - Therefore, **Carboxylic Acids ($pK_a \approx 4 - 5$) are 10^6 times more acidic than Phenols ($pK_a \approx 10$)**, and far more acidic than alcohols ($pK_a \approx 16 - 18$).

2. Substituent Inductive & Resonance Effects on Acidity (pK_a)

- **Electron-Withdrawing Groups ($-I, -M$):** Disperse the negative charge of the carboxylate ion, stabilizing it and **increasing acidity** (lowering pK_a).
  - Order of $-I$ power: $CF_3 > NO_2 > CN > F > Cl > Br > I > C_6H_5$.
- **Electron-Donating Groups ($+I$):** Intensify the negative charge on the carboxylate ion, destabilizing it and **decreasing acidity** (raising pK_a).
- **Inductive Distance Effect:** The closer the electron-withdrawing substituent to the $-COOH$ group, the greater the acid strengthening (α-chlorobutanoic acid > β-chlorobutanoic acid > γ-chlorobutanoic acid).
- **Sacred CUET Acidity Hierarchy:**

$$\text{OH} > CHCl_2\ COOH > NO_2CH_2\ COOH > NC\text{-}CH_2\ COOH > FCH_2\ COOH > ClCH_2\ COOH > BrCH_2\ COOH > HCOOH >$$

  - Formic acid (HCOOH, $pK_a = 3.75$) is more acidic than acetic acid ($CH_3\ COOH$, $pK_a = 4.76$) because the methyl group exerts an electron-releasing $+I$ effect.

- Benzoic acid (C_6H_5 COOH, $pK_a = 4.20$) is stronger than acetic acid due to the $-I$ effect of sp^2 phenyl ring, but weaker than formic acid.

3. Preparative Methods for Carboxylic Acids

1. **Oxidation of Primary Alcohols & Aldehydes:** Using alkaline potassium permanganate ($KMnO_4$) followed by acidification.
2. **Hydrolysis of Nitriles & Amides:**

$$R - CN + H_2O \xrightarrow{H^+} R - CONH_2 \xrightarrow{H^+} R - COOH + NH_4^+$$

3. **Carboxylation of Grignard Reagents (Chain Extension by One Carbon):**
   - Grignard reagents add across the $C = O$ double bond of solid carbon dioxide (dry ice) to form a magnesium carboxylate complex, which on acidic hydrolysis yields a carboxylic acid:

$$R - MgX + O{=}C{=}O \rightarrow R - COOMgX \xrightarrow{H_3O^+} R - COOH + Mg(OH)X$$

4. Chemical Reactions of Carboxylic Acids

1. **Distinction from Phenol via Sodium Bicarbonate ($NaHCO_3$):**
   - Carboxylic acids react vigorously with aqueous $NaHCO_3$ releasing brisk effervescence of **Carbon Dioxide ($CO_2 \uparrow$)**:

$$R - COOH + NaHCO_3 \rightarrow R - COONa + H_2O + CO_2 \uparrow$$

   - Phenols (except 2,4,6-trinitrophenol / picric acid) do NOT release CO_2 from $NaHCO_3$.
2. **Derivative Formations:**
   - **Acid Chlorides:** $R - COOH + SOCl_2 \rightarrow R - COCl + SO_2 \uparrow + HCl \uparrow$ (Thionyl chloride is preferred because gaseous by-products escape).
   - **Anhydrides:** Heating two molecules of acid with phosphorus pentoxide (P_2O_5) eliminates water.
   - **Esters (Fischer Esterification):** Reaction with alcohols in the presence of concentrated H_2SO_4.
3. **Hell-Volhard-Zelinsky (HVZ) Reaction (α-Halogenation):**
   - Carboxylic acids having α-hydrogens react with chlorine or bromine in the presence of a small amount of **red phosphorus** to form **α-halocarboxylic acids**:

$$R - CH_2 - COOH + X_2 \xrightarrow{\text{red P}} R - CH(X) - COOH \quad (X = Cl, Br)$$

   - **Substrate Constraint:** Formic acid and benzoic acid do not give the HVZ reaction because they lack α-hydrogens!
4. **Decarboxylation with Soda-Lime:**
   - Heating sodium salts of carboxylic acids with **Soda-Lime (NaOH + CaO in 3 : 1 ratio)** eliminates carbon dioxide to produce an alkane with **one fewer carbon atom**:

$$R - COONa + NaOH \xrightarrow{CaO,\Delta} R - H + Na_2CO_3$$

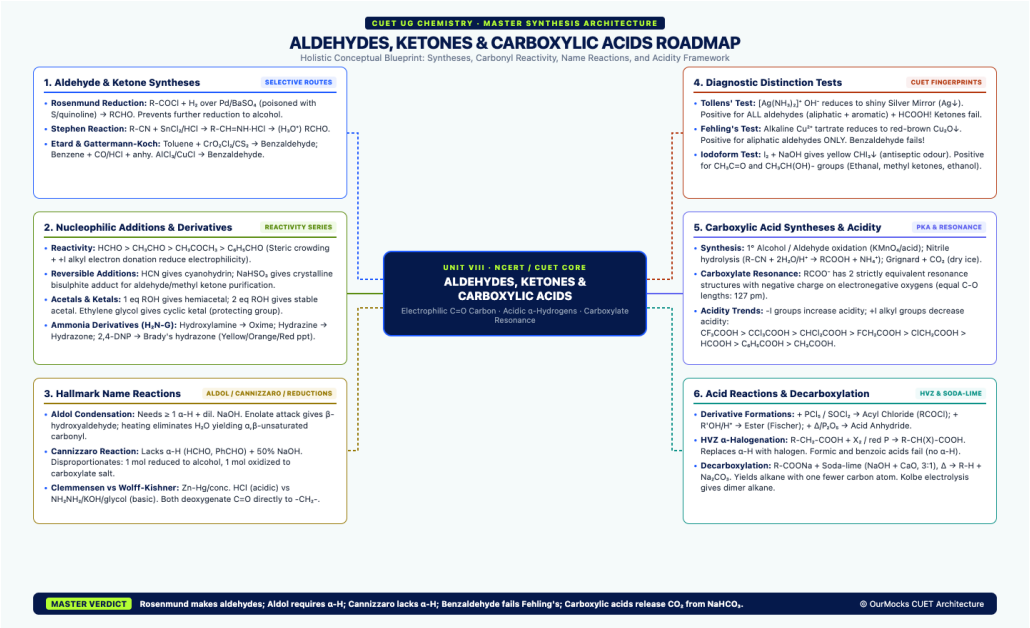


Figure 3: Master Synthesis Architecture: Complete Unit VIII cognitive roadmap integrating aldehyde/ketone preparations, nucleophilic additions, name reactions, diagnostic tests, and carboxylic acid acidity.

UNIT VIII MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Reagent / Name Reaction	Key Formulation / Intermediate	CUET Problem Strategy & Trap
Rosenmund Reduction	$R-\mathrm{COCl}+\mathrm{H_2}\xrightarrow{\mathrm{Pd\ /\ BaSO_4/\ quinoline}}R-\mathrm{CHO}$	$\mathrm{BaSO_4}$ poisons catalyst; prevents over-reduction. HCHO cannot be prepared.
Stephen Reaction	$R-\mathrm{CN}+\mathrm{SnCl_2/\ HCl}\rightarrow R-\mathrm{CH{=}NH}\xrightarrow{\mathrm{H_3O^+}}R-\mathrm{CHO}$	Selective reduction of nitriles to aldehydes via aldimine hydrochloride.
Etard Reaction	$C_6H_5-\mathrm{CH_3}+\mathrm{CrO_2Cl_2}\rightarrow$ Chromium complex $\rightarrow \mathrm{C_6H_5\ CHO}$	Chromyl chloride selectively oxidises methyl side chain to benzaldehyde.
Gattermann-Koch	$\mathrm{C_6H_6}+\mathrm{CO}+\mathrm{HCl}\xrightarrow{\mathrm{anhy.\ AlCl_3/\ CuCl}}\mathrm{C_6H_5\ CHO}$	Direct formylation of benzene ring with carbon monoxide and HCl under pressure.
Dialkylcadmium (R_2 Cd)	$2R'-\mathrm{COCl}+R_2\ \mathrm{Cd}\rightarrow 2R'-\mathrm{CO}-R+\mathrm{CdCl_2}$	Stops selectively at ketone; less nucleophilic than Grignard, never forms 3° alcohol.
Alkyne Hydration	$\mathrm{HC\equiv CH}+\mathrm{H_2\ O}\xrightarrow{\mathrm{Hg^{2+}/H^+}}\mathrm{CH_3\ CHO}$; other alkynes give ketones	Acetylene gives ethanal; propyne gives acetone via Markovnikov enol tautomerism.
Nucleophilic Addition	$\mathrm{HCHO} > R\mathrm{CHO} > R_2\ \mathrm{CO} > \mathrm{ArCHO}$	Steric crowding and $+I$ alkyl groups retard nucleophilic attack at carbonyl carbon.
Semicarbazide Test	$>C=O+\mathrm{H_2\ N-CO-NH-NH_2}\rightarrow >C=N-\mathrm{NH-CO-NH_2}$	Only terminal hydrazine $-\mathrm{NH_2}$ reacts; amide $-\mathrm{NH_2}$ is unreactive due to resonance.

Core Reagent / Name Reaction	Key Formulation / Intermediate	CUET Problem Strategy & Trap
Aldol Condensation	$2CH_3\ CHO \xrightarrow{\text{dil. NaOH}} \text{Aldol} \xrightarrow{\Delta}$ But-2-enal	Strictly requires ≥ 1 α-hydrogen. Forms α,β-unsaturated carbonyl on heating.
Cannizzaro Reaction	$2\ HCHO + 50\%\ NaOH \rightarrow CH_3\ OH + HCOONa$	Requires zero α-H. Disproportionation; hydride transfer is rate-determining step.
Clemmensen Reduction	$> C = O \xrightarrow{\text{Zn-Hg / conc. HCl}} > CH_2$	Acidic conditions; avoid if molecule contains acid-sensitive groups like $-OH$.
Wolff-Kishner	$> C = O \xrightarrow{NH_2NH_2/\ \text{KOH, glycol},\Delta} > CH_2 + N_2 \uparrow$	Basic conditions; avoid if molecule contains base-sensitive groups like esters or halides.
Tollens' Test	All aldehydes $+\left[Ag(NH_3)_2\right]^+ \rightarrow$ Ag mirror	Distinguishes aldehydes from ketones. Formic acid gives positive test!
Fehling's Test	Aliphatic aldehydes $+Cu^{2+} \rightarrow Cu_2\ O \downarrow$ (Red-brown)	Aromatic aldehydes (Benzaldehyde) FAIL Fehling's test completely.
Iodoform Test	$CH_3 - C{=}O$ or $CH_3 - CH(OH)\text{-} + I_2/\ NaOH \rightarrow CHI_3 \downarrow$	Yellow precipitate with antiseptic odour. Ethanal is the ONLY aldehyde passing.
Carboxylate Resonance	$R - C(=O) - O^- \longleftrightarrow R - C(\text{-}\ O^-) = O$	Two equivalent canonical structures with charge on oxygen. Far more acidic than phenol.
Acidity Hierarchy	$CF_3\ COOH > CCl_3\ COOH > ClCH_2\ COOH > HCOOH > CH_3\ COOH$	$-I$ electron-withdrawing groups increase acidity; $+I$ alkyl groups decrease acidity.
HVZ Reaction	$R - CH_2 - COOH + \frac{X_2}{\text{red P}} \rightarrow R - CH(X) - COOH$	α-Halogenation of carboxylic acids. Formic and benzoic acids fail (no α -H).

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. Which of the following organic carbonyl compounds yields a positive silver mirror with Tollens' reagent but FAILS to reduce Fehling's solution to red-brown cuprous oxide?

(A) Ethanal (Acetaldehyde) **(B)** Propanal (Propionaldehyde)
(C) Benzaldehyde **(D)** Propan-2-one (Acetone)

Sol. Q1. Tollens' reagent is a mild oxidising agent that oxidises both aliphatic and aromatic aldehydes to carboxylate ions with deposition of shiny metallic silver mirror. In contrast, Fehling's solution is a weaker oxidising agent that oxidises aliphatic aldehydes only. In benzaldehyde (C_6H_5 CHO), the carbonyl carbon is resonance-stabilized by conjugation with the electron-rich aromatic ring, reducing its electrophilic character so that it cannot be oxidised by Fehling's solution. **Answer: (C)**

Q2. In the Rosenmund reduction for preparing aldehydes from acid chlorides, barium sulphate ($BaSO_4$) poisoned with sulphur or quinoline is added to the palladium catalyst primarily to:

(A) Accelerate the rate of catalytic hydrogenation **(B)** Prevent over-reduction of the aldehyde product to primary alcohol
(C) Facilitate the formation of formaldehyde **(D)** Increase the solubility of the acyl chloride in boiling xylene

Sol. Q2. Acyl chlorides are exceptionally reactive. Unpoisoned palladium catalyst rapidly reduces the acid chloride to an aldehyde and immediately reduces the aldehyde further to a primary alcohol. Adding $BaSO_4$ along with sulphur or quinoline acts as a catalyst poison (moderator) that reduces the catalytic efficiency of palladium just enough so that reduction stops selectively at the aldehyde stage. **Answer: (B)**

Q3. Which of the following pairs of carbonyl compounds will undergo the Cannizzaro reaction when heated with concentrated (50%) aqueous sodium hydroxide?

(A) Ethanal and Propanal **(B)** Methanal (Formaldehyde) and Benzaldehyde
(C) Propan-2-one and Butan-2-one **(D)** Ethanal and Benzaldehyde

Sol. Q3. The Cannizzaro reaction is a characteristic disproportionation reaction exhibited strictly by aldehydes that lack α-hydrogen atoms. Both methanal (HCHO) and benzaldehyde (C_6H_5 CHO) contain no hydrogen on the carbon adjacent to the carbonyl group and therefore undergo Cannizzaro reaction when treated with 50% NaOH. Ethanal possesses three α-hydrogens and undergoes aldol condensation instead. **Answer: (B)**

Q4. The yellow crystalline precipitate obtained during a positive iodoform test with iodine and sodium hydroxide has the chemical formula:

(A) CH_2I_2 **(B)** CH_3I
(C) CHI_3 **(D)** CI_4

Sol. Q4. The iodoform test utilizes iodine (I_2) and aqueous NaOH (forming sodium hypoiodite NaOI). Carbonyl compounds containing a methyl ketone unit ($CH_3-C{=}O$) or alcohols containing a methyl carbinol unit ($CH_3-CH(OH)$-) undergo triiodination at the α-methyl carbon followed by nucleophilic cleavage by hydroxide to produce triiodomethane (**Iodoform, CHI_3**), which precipitates as bright yellow crystals (melting point 119° C) with a pungent antiseptic odour. **Answer: (C)**

Q5. To synthesize a ketone cleanly from an acid chloride without risking over-addition to a tertiary alcohol, the preferred organometallic reagent is:

(A) A Grignard reagent ($R-MgX$) **(B)** An organolithium reagent ($R-Li$)
(C) A dialkylcadmium reagent (R_2Cd) **(D)** Sodium borohydride ($NaBH_4$)

Sol. Q5. Grignard reagents ($R-MgX$) and organolithium reagents ($R-Li$) are intensely nucleophilic; when they react with acyl chlorides, the initially formed ketone is even more electrophilic and rapidly undergoes a second addition to yield a tertiary alcohol. In contrast, dialkylcadmium (R_2Cd) is far less nucleophilic (owing to the higher electronegativity and covalent character of cadmium); it reacts smoothly with acyl chlorides to yield ketones and stops completely without reacting with the ketone product. **Answer: (C)**

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. Which of the following represents the correct decreasing order of acidic strength among the given carboxylic acids?

(A) $CF_3\ COOH > CCl_3\ COOH > CHCl_2\ COOH > CH_2\ ClCOOH$
(B) $CH_3\ COOH > CH_2\ ClCOOH > CHCl_2\ COOH > CCl_3\ COOH$
(C) $CH_2\ FCOOH < CH_2\ ClCOOH < CH_2\ BrCOOH < CH_2\ ICOOH$
(D) $CH_3\ COOH > HCOOH > C_6H_5\ COOH > CF_3\ COOH$

Sol. Q6. Acidic strength of carboxylic acids depends directly on the stabilization of the conjugate carboxylate anion ($R-COO^-$). Electron-withdrawing groups ($-I$ inductive effect) disperse the negative charge across the carboxylate oxygen atoms, heavily stabilizing the conjugate base and increasing acidity. Three fluorine atoms in trifluoroacetic acid exert an extraordinary $-I$ pull, followed by trichloroacetic, dichloroacetic, and monochloroacetic acid: $CF_3\ COOH > CCl_3\ COOH > CHCl_2\ COOH > CH_2\ ClCOOH$. **Answer: (A)**

Q7. Carboxylic acids possess substantially higher boiling points than alcohols and aldehydes of comparable molecular masses primarily because:

(A) Carboxylic acids possess stronger covalent bonds
(B) Carboxylic acid molecules associate into stable cyclic dimers via two intermolecular hydrogen bonds
(C) The carbonyl carbon is sp^2-hybridised
(D) Carboxylic acids undergo spontaneous intramolecular ionization in liquid state

Sol. Q7. Carboxylic acids exhibit exceptionally high boiling points because each carboxyl group contains both a hydrogen-bond donor (the acidic $O-H$ proton) and a hydrogen-bond acceptor (the carbonyl oxygen). In liquid and even vapor phase, two molecules of carboxylic acid interlock via **two intermolecular hydrogen bonds** to form a stable eight-membered cyclic dimer. This effectively doubles the molecular size and demands substantial thermal energy to disrupt. **Answer: (B)**

Q8. The Hell-Volhard-Zelinsky (HVZ) reaction transforms propanoic acid ($CH_3CH_2\ COOH$) into 2-bromopropanoic acid when treated with:

(A) Bromine in aqueous sodium hydroxide at room temperature
(B) Bromine in the presence of red phosphorus ($\frac{Br_2}{\text{red P}}$)
(C) Hydrobromic acid (HBr) under reflux
(D) Phosphorus tribromide (PBr_3) followed by hydrogen peroxide

Sol. Q8. The HVZ reaction is specific for the α-halogenation of aliphatic carboxylic acids containing α-hydrogen atoms. Red phosphorus reacts with halogen in situ to form phosphorus trihalide (PBr_3), which converts a portion of the acid into acyl halide. The acyl halide rapidly enolises and brominates exclusively at the α-carbon, followed by halogen exchange with unreacted acid to yield the **α-bromocarboxylic acid** ($CH_3-CH(Br)$-COOH). **Answer: (B)**

Q9. Heating anhydrous sodium ethanoate ($CH_3\ COONa$) with soda-lime ($NaOH + CaO$) at elevated temperature yields:

(A) Ethane (C_2H_6)
(B) Methane (CH_4)
(C) Ethanol ($C_2H_5\ OH$)
(D) Ethanal ($CH_3\ CHO$)

Sol. Q9. Decarboxylation of the sodium salt of a carboxylic acid with soda-lime (a 3 : 1 fused mixture of NaOH and CaO) eliminates carbon dioxide as sodium carbonate, yielding an alkane containing **one carbon atom fewer** than the parent carboxylate salt: $CH_3\ COONa + NaOH \xrightarrow{CaO,\Delta} CH_4 + Na_2CO_3$. Calcium oxide (CaO) acts as a drying agent, keeping the reaction mixture porous and preventing glass corrosion. **Answer: (B)**

Q10. When benzene is treated with carbon monoxide (CO) and hydrogen chloride (HCl) gas in the presence of anhydrous aluminium chloride and cuprous chloride, the product obtained is:

(A) Benzoic acid
(B) Benzaldehyde
(C) Acetophenone
(D) Benzyl alcohol

Sol. Q10. This is the **Gattermann-Koch reaction**. Carbon monoxide and HCl react under high pressure in the presence of anhydrous $AlCl_3$ and CuCl to generate formyl chloride (HCOCl) in situ, which acts as the electrophile in an electrophilic aromatic substitution to install a formyl ($-CHO$) group directly onto the benzene ring, producing **Benzaldehyde**. Acetophenone requires acetyl chloride in Friedel-Crafts acylation. **Answer: (B)**

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. Which of the following reagents can cleanly reduce butan-2-one directly to butane under basic reaction conditions without affecting an ester group if present?

(A) Amalgamated zinc and concentrated hydrochloric acid (Zn-Hg / conc. HCl)
(B) Hydrazine followed by heating with potassium hydroxide in ethylene glycol ($NH_2 \frac{NH_2}{KOH}, \Delta$)
(C) Lithium aluminium hydride ($LiAlH_4$) in dry ether
(D) Sodium borohydride ($NaBH_4$) in ethanol

Sol. Q11. Both Clemmensen and Wolff-Kishner reductions reduce carbonyl groups ($> C = O$) directly to methylene ($-CH_2-$). However, Clemmensen uses concentrated mineral acid (HCl) which hydrolyses esters and acid-sensitive groups. Wolff-Kishner reduction employs **hydrazine followed by KOH in ethylene glycol at 453 — 473 K**, providing strongly basic conditions suitable for deoxygenating carbonyls to alkanes without acidic cleavage. **Answer: (B)**

Q12. When semicarbazide ($H_2\ N-CO-NH-NH_2$) reacts with an aldehyde to form a semicarbazone, which nitrogen atom functions as the active nucleophile?

(A) The amide nitrogen atom attached directly to the carbonyl group ($-CO-NH_2$)
(B) The terminal hydrazine nitrogen atom ($-NH-NH_2$)
(C) The secondary nitrogen atom ($-NH-CO-$)
(D) Both nitrogen atoms react simultaneously to form a cyclic diazole

Sol. Q12. In semicarbazide, the lone pair on the amide nitrogen atom ($-CO-NH_2$) is delocalized into the adjacent carbonyl group via resonance ($O{=}C-NH_2 \overset{-O}{\longleftrightarrow} C=NH_2^+$), rendering it strongly electron-deficient and non-nucleophilic. In contrast, the terminal hydrazine nitrogen atom ($-NH-NH_2$) is not in conjugation with the carbonyl group; its lone pair is fully localized, making it exclusively nucleophilic to attack the aldehyde. **Answer: (B)**

Q13. Which of the following carboxylic acids releases brisk effervescence of carbon dioxide (CO_2) when treated with aqueous sodium hydrogen carbonate ($NaHCO_3$), and also gives a positive silver mirror with Tollens' reagent?

(A) Acetic acid ($CH_3\ COOH$)
(B) Formic acid (Methanoic acid, HCOOH)
(C) Benzoic acid ($C_6H_5\ COOH$)
(D) Oxalic acid (HOOC-COOH)

Sol. Q13. Formic acid (HCOOH) possesses a unique dual structure: viewing from one end it is a carboxylic acid ($-COOH$), allowing it to release brisk effervescence of CO_2 from $NaHCO_3$. Viewing from the other end, the carbonyl carbon carries a direct hydrogen ($H-C{=}O$), conferring the reducing properties of an aldehyde. Therefore, formic acid uniquely reduces Tollens' reagent ($Ag^+ \rightarrow Ag$) and Fehling's solution, while acetic and benzoic acids do not. **Answer: (B)**

Q14. In a cross-Cannizzaro reaction between benzaldehyde ($C_6H_5\ CHO$) and formaldehyde (HCHO) in the presence of concentrated sodium hydroxide, the major products formed are:

(A) Sodium benzoate and Methanol
(B) Benzyl alcohol and Sodium formate
(C) Benzoic acid and Formic acid
(D) Benzyl alcohol and Methanol

Sol. Q14. Formaldehyde (HCHO) lacks alkyl or aryl groups; its carbonyl carbon is far less sterically hindered and significantly more electrophilic than that of benzaldehyde. Consequently, nucleophilic hydroxide ion (OH^-) attacks formaldehyde much more rapidly. The resulting intermediate transfers a hydride ion ($:H^-$) to the less reactive benzaldehyde. Thus, **formaldehyde is exclusively oxidised to sodium formate (HCOONa)** while **benzaldehyde is exclusively reduced to benzyl alcohol ($C_6H_5CH_2\ OH$)**. **Answer: (B)**

Q15. Treatment of propan-2-one (acetone) with dry hydrogen chloride gas in the presence of excess dry ethanol yields which of the following compounds?

(A) A stable diethyl ketal / acetal
(B) A hemiacetal that resists further reaction
(C) A cyclic glycol ketal
(D) Propan-2-ol and ethyl acetate

Sol. Q15. Ketones react reluctantly with monohydric alcohols; however, treatment of propan-2-one with excess dry monohydric alcohol in the presence of anhydrous HCl gas can yield a ketal (2,2-diethoxypropane). More importantly, for synthetic protection in organic chemistry, ketones react with **1,2-diols (ethylene glycol)** in the presence of dry HCl or p-toluenesulphonic acid to form highly stable **cyclic ketals (ethylene glycol ketals)**, which are immune to basic reagents. **Answer: (A)**

© OurMocks · www.ourmocks.com · CUET UG Chemistry · Unit VIII: Aldehydes, Ketones and Carboxylic Acids · 2027 Examination Baseline